AWS Academy Cloud Foundations
Module 08 Student Guide
Version 2.0.19

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Welcome to Module 8: Databases

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Module overview

Topics

- Amazon Relational Database Service (Amazon RDS)
- Amazon DynamoDB
- Amazon Redshift
- Amazon Aurora

Demos

- Amazon RDS console
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- Lab 5: Build Your DB Server and Interact with Your DB Using an App

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The business world is constantly changing and evolving. By accurately recording, updating, and tracking data on an efficient and regular basis, companies can use the immense potential from the insights that they obtain from their data. Database management systems are the crucial link for managing this data. Like other cloud services, cloud databases offer significant cost advantages over traditional database strategies.

In this module, you will learn about Amazon Relational Database Service (or Amazon RDS), Amazon DynamoDB, Amazon Redshift, and Amazon Aurora.

This module will address the following topics:
- Amazon Relational Database Service (Amazon RDS)
- Amazon DynamoDB
- Amazon Redshift
- Amazon Aurora

The module includes two recorded demonstrations that will show you how to access and interact with Amazon RDS and Amazon DynamoDB by using the AWS Management Console.

The module also includes a hands-on lab where you will set up an Amazon RDS database solution.

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The module also includes an activity that challenges you to select the appropriate database service for a business case.

Finally, you will be asked to complete a knowledge check that will test your understanding of the key concepts that are covered in this module.

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Module objectives

After completing this module, you should be able to:

- Explain Amazon Relational Database Service (Amazon RDS)
- Identify the functionality in Amazon RDS
- Explain Amazon DynamoDB
- Identify the functionality in Amazon DynamoDB
- Explain Amazon Redshift
- Explain Amazon Aurora
- Perform tasks in an RDS database, such as launching, configuring, and interacting

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In this module, you will learn about key concepts that are related to database solutions, including:

- Understanding the different database services in the cloud.
- Discovering the differences between unmanaged and managed database solutions.
- Understanding the differences between Structured Query Language (or SQL) and NoSQL databases.
- Comparing the availability differences of alternative database solutions.

The goal of this module is to help you understand the database resources that are available to power your solution. You will also review the different service features that are available, so you can begin to understand how different choices impact things like solution availability

After completing this module, you should be able to:

- Explain Amazon Relational Database Service (Amazon RDS)
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- Identify the functionality in Amazon DynamoDB
- Explain Amazon Redshift
- Explain Amazon Aurora
- Perform tasks in an RDS database, such as launching, configuring, and interacting

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Introducing Section 1: Amazon Relational Database Service.

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Amazon Relational Database Service

Amazon Relational Database Service (Amazon RDS)

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Welcome to an introduction to the foundational database services that are available on Amazon Web Services (AWS). This module begins with Amazon Relational Database Service (Amazon RDS).

This section starts by reviewing the differences between a managed and unmanaged service in relation to Amazon RDS.

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Unmanaged versus managed services

Unmanaged:

Scaling, fault tolerance, and availability are managed by you.

Managed:

Scaling, fault tolerance, and availability are typically built into the service.

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AWS solutions typically fall into one of two categories: unmanaged or managed.

Unmanaged services are typically provisioned in discrete portions as specified by the user. You must manage how the service responds to changes in load, errors, and situations where resources become unavailable. Say that you launch a web server on an Amazon Elastic Compute Cloud (Amazon EC2) instance. Because Amazon EC2 is an unmanaged solution, that web server will not scale to handle increased traffic load or replace unhealthy instances with healthy ones unless you specify that it use a scaling solution, such as AWS Automatic Scaling. The benefit to using an unmanaged service is that you have more fine-tuned control over how your solution handles changes in load, errors, and situations where resources become unavailable.

Managed services require the user to configure them. For example, you create an Amazon Simple Storage Service (Amazon S3) bucket and then set permissions for it. However, managed services typically require less configuration. Say that you have a static website that you host in a cloud-based storage solution, such as Amazon S3. The static website does not have a web server. However, because Amazon S3 is a managed solution, features such as scaling, fault-tolerance, and availability would be handled automatically and internally by Amazon S3.

Now, you will look at the challenges of running an unmanaged, standalone relational database. Then, you will learn how Amazon RDS addresses these challenges.

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Challenges of relational databases

- Server maintenance and energy footprint
- Software installation and patches
- Database backups and high availability
- Limits on scalability
- Data security
- Operating system (OS) installation and patches

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When you run your own relational database, you are responsible for several administrative tasks, such as server maintenance and energy footprint, software, installation and patching, and database backups. You are also responsible for ensuring high availability, planning for scalability, data security, and operating system (OS) installation and patching. All these tasks take resources from other items on your to-do list, and require expertise in several areas.

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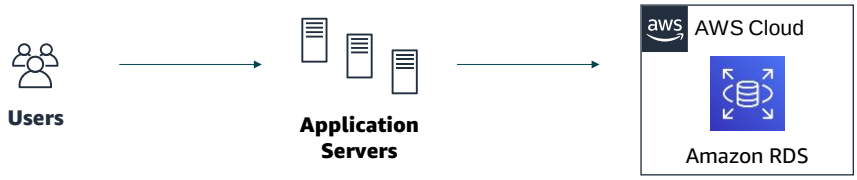
Amazon RDS

Managed service that sets up and operates a relational database in the cloud.



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Amazon RDS is a managed service that sets up and operates a relational database in the cloud.

To address the challenges of running an unmanaged, standalone relational database, AWS provides a service that sets up, operates, and scales the relational database without any ongoing administration. Amazon RDS provides cost-efficient and resizable capacity, while automating time-consuming administrative tasks.

Amazon RDS enables you to focus on your application, so you can give applications the performance, high availability, security, and compatibility that they need. With Amazon RDS, your primary focus is your data and optimizing your application.

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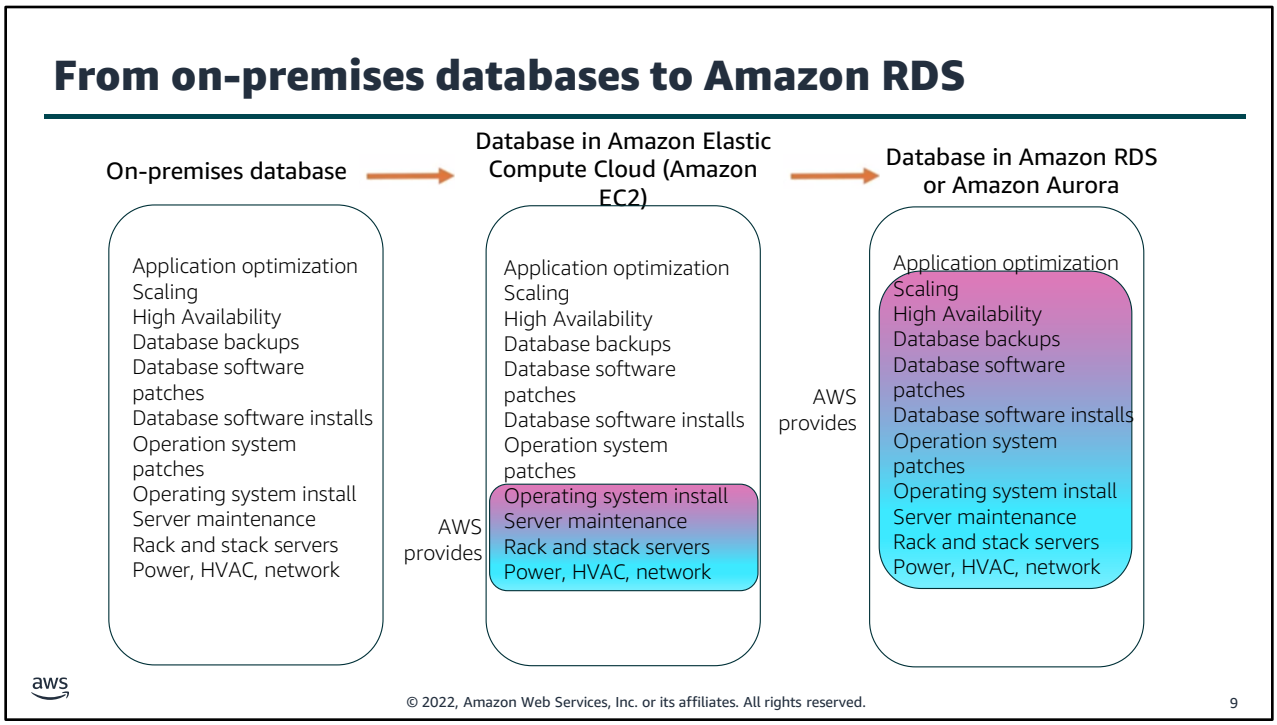

What does the term **managed services** mean?

When your database is on premises, the database administrator is responsible for everything. Database administration tasks include optimizing applications and queries; setting up the hardware; patching the hardware; setting up networking and power; and managing heating, ventilation, and air conditioning (HVAC).

If you move to a database that runs on an **Amazon Elastic Compute Cloud (Amazon EC2) instance**, you no longer need to manage the underlying hardware or handle data center operations. However, you are still responsible for patching the OS and handling all software and backup operations.

If you set up your database on **Amazon RDS** or **Amazon Aurora**, you reduce your administrative responsibilities. By moving to the cloud, you can automatically scale your database, enable high availability, manage backups, and perform patching. Thus, you can focus on what really matters most—optimizing your application.

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Managed services responsibilities

You manage:

- Application optimization

AWS manages:

- OS installation and patches
- Database software installation and patches
- Database backups
- High availability
- Scaling
- Power and racking and stacking servers
- Server maintenance

Amazon RDS

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With Amazon RDS, you manage your application optimization. AWS manages installing and patching the operating system, installing and patching the database software, automatic backups, and high availability.

AWS also scales resources, manages power and servers, and performs maintenance.

Offloading these operations to the managed Amazon RDS service reduces your operational workload and the costs that are associated with your relational database. You will now go through a brief overview of the service and a few potential use cases.

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Amazon RDS DB instances

Amazon RDS

Amazon RDS DB main instance

DB Instance Class

- CPU
- Memory
- Network performance

DB Instance Storage

- Magnetic
- General Purpose (solid state drive, or SSD)
- Provisioned IOPS

MySQL

IBM DB2

Microsoft SQL Server

PostgreSQL

MariaDB

Oracle

DB engines

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The basic building block of Amazon RDS is the database instance. A **database instance** is an isolated database environment that can contain multiple user-created databases. It can be accessed by using the same tools and applications that you use with a standalone database instance. The resources in a database instance are determined by its database instance class, and the type of storage is dictated by the type of disks.

Database instances and storage differ in performance characteristics and price, which enable you to customize your performance and cost to the needs of your database. When you choose to create a database instance, you must first specify which database engine to run. Amazon RDS currently supports six databases: MySQL, IBM DB2, Microsoft SQL Server, PostgreSQL, MariaDB, and Oracle.

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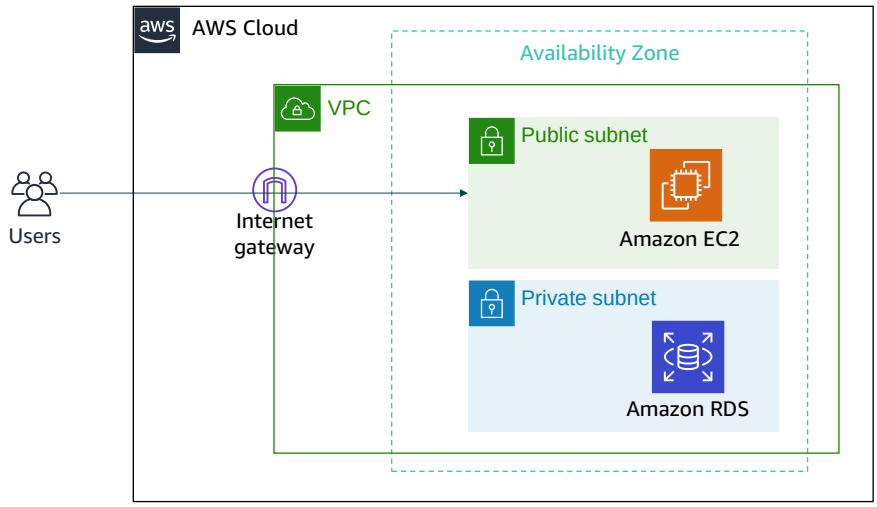
Amazon RDS in a virtual private cloud (VPC)



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You can run an instance by using **Amazon Virtual Private Cloud (Amazon VPC).** When you use a virtual private cloud (VPC), you have control over your virtual networking environment.

You can select your own IP address range, create subnets, and configure routing and access control lists (ACLs). The basic functionality of Amazon RDS is the same whether or not it runs in a VPC. Usually, the database instance is isolated in a private subnet and is only made directly accessible to indicated application instances. Subnets in a VPC are associated with a single Availability Zone, so when you select the subnet, you are also choosing the Availability Zone (or physical location) for your database instance.

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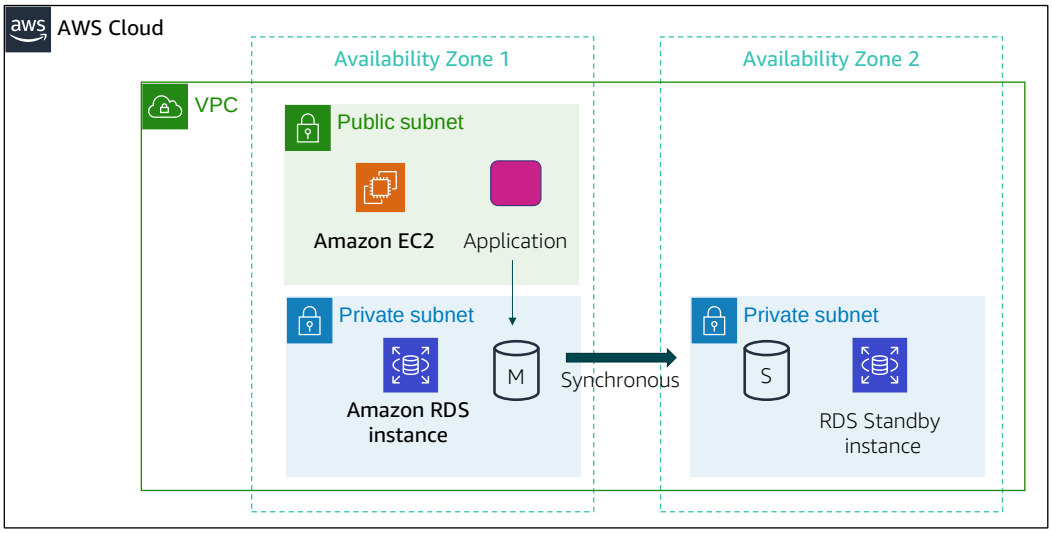
High availability with Multi-AZ deployment (1 of 2)



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One of the most powerful features of Amazon RDS is the ability to configure your database instance for high availability with a Multi-AZ deployment. After a Multi-AZ deployment is configured, Amazon RDS automatically generates a standby copy of the database instance in another Availability Zone within the same VPC. After seeding the database copy, transactions are synchronously replicated to the standby copy. Running a database instance in a Multi-AZ deployment can enhance availability during planned system maintenance, and it can help protect your databases against database instance failure and Availability Zone disruption.

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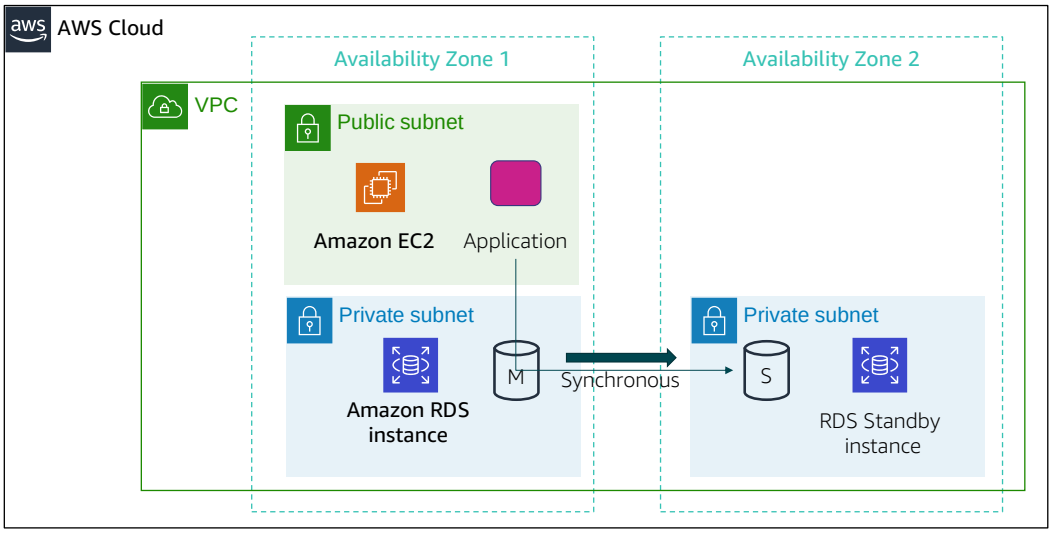
High availability with Multi-AZ deployment (2 of 2)



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Therefore, if the main database instance fails in a Multi-AZ deployment, Amazon RDS automatically brings the standby database instance online as the new main instance. The synchronous replication minimizes the potential for data loss. Because your applications reference the database by name by using the Amazon RDS Domain Name System (DNS) endpoint, you don't need to change anything in your application code to use the standby copy for failover.

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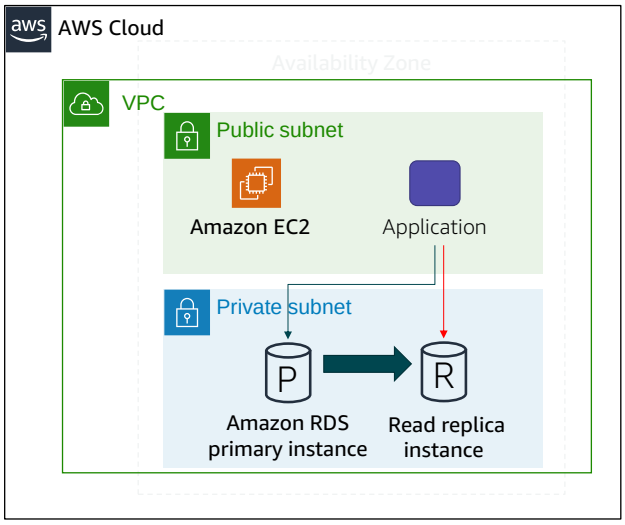
Amazon RDS read replicas

Features

- Offers asynchronous replication
- Can be promoted to primary if needed

Functionality

- Use for read-heavy database workloads
- Offload read queries



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Amazon RDS also supports the creation of read replicas for MySQL, MariaDB, PostgreSQL, and Amazon Aurora. Updates that are made to the source database instance are asynchronously copied to the read replica instance. You can reduce the load on your source database instance by routing read queries from your applications to the read replica. Using read replicas, you can also scale out beyond the capacity constraints of a single database instance for read-heavy database workloads. Read replicas can also be promoted to become the primary database instance, but this requires manual action because of asynchronous replication.

Read replicas can be created in a different Region than the primary database. This feature can help satisfy disaster recovery requirements or reduce latency by directing reads to a read replica that is closer to the user.

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Use cases

Web and mobile applications	✓ High throughput ✓ Massive storage scalability ✓ High availability
Ecommerce applications	✓ Low-cost database ✓ Data security ✓ Fully managed solution
Mobile and online games	✓ Rapidly grow capacity ✓ Automatic scaling ✓ Database monitoring

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Amazon RDS works well for web and mobile applications that need a database with high throughput, massive storage scalability, and high availability. Because Amazon RDS does not have any licensing constraints, it fits the variable usage pattern of these applications. For small and large ecommerce businesses, Amazon RDS provides a flexible, secure, and low-cost database solution for online sales and retailing. Mobile and online games require a database platform with high throughput and availability. Amazon RDS manages the database infrastructure, so game developers do not need to worry about provisioning, scaling, or monitoring database servers.

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When to Use Amazon RDS

Use Amazon RDS when your application requires:

- Complex transactions or complex queries
- A medium to high query or write rate – Up to 30,000 IOPS (15,000 reads + 15,000 writes)
- No more than a single worker node or shard
- High durability

- **Do not use Amazon RDS when your application requires:**
  - Massive read/write rates (for example, 150,000 write/second)
  - Sharding due to high data size or throughput demands
  - Simple GET or PUT requests and queries that a NoSQL database can handle
  - Relational database management system (RDBMS) customization

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Use Amazon RDS when your application requires:
- Complex transactions or complex queries
- A medium to high query or write rate – up to 30,000 IOPS (15,000 reads + 15,000 writes)
- No more than a single worker node or shard
- High durability

Do not use Amazon RDS when your application requires:
- Massive read/write rates (for example 150,000 writes per second)
- Sharding due to high data size or throughput demands
- Simple GET or PUT requests and queries that a NoSQL database can handle
- Or, relational database management system (RDBMS) customization

For circumstances when you should not use Amazon RDS, consider either using a NoSQL database solution (such as DynamoDB) or running your relational database engine on Amazon EC2 instances instead of Amazon RDS (which will provide you with more options for customizing your database).

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Amazon RDS: Clock-hour billing and database characteristics

Clock-hour billing –

- Resources incur charges when running

Database characteristics –

- Physical capacity of database:
  - Engine
  - Size
  - Memory class

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When you begin to estimate the cost of Amazon RDS, you must consider the clock hours of service time, which are resources that incur charges when they are running (for example, from the time you launch a database instance until you terminate the instance).

Database characteristics should also be considered. The physical capacity of the database you choose will affect how much you are charged. Database characteristics vary depending on the database engine, size, and memory class.

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Amazon RDS: DB purchase type and multiple DB instances

DB purchase type –

- On-Demand Instances
  - Compute capacity by the hour
- Reserved Instances
  - Low, one-time, upfront payment for database instances that are reserved with a 1-year or 3-year term

Number of DB instances –

- Provision multiple DB instances to handle peak loads

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Consider the database purchase type. When you use On-Demand Instances, you pay for compute capacity for each hour that your database instance runs, with no required minimum commitments. With Reserved Instances, you can make a low, one-time, upfront payment for each database instance you want to reserve for a 1-year or 3-year term.

Also, you must consider the number of database instances. With Amazon RDS, you can provision multiple database instances to handle peak loads.

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Amazon RDS: Storage

Provisioned storage –

- No charge
  - Backup storage of up to 100 percent of database storage for an active database
- Charge (*GB/month*)
  - Backup storage for terminated DB instances

Additional storage –

- Charge (*GB/month*)
  - Backup storage in addition to provisioned storage

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Consider provisioned storage. There is no additional charge for backup storage of up to 100 percent of your provisioned database storage for an active database instance. After the database instance is terminated, backup storage is billed per GB, per month.

Also consider the amount of backup storage in addition to the provisioned storage amount, which is billed per GB, per month.

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Amazon RDS: Deployment type and data transfer

Requests –

- The number of input and output requests that are made to the database

Deployment type—Storage and I/0 charges vary, depending on whether you deploy to –

- Single Availability Zone
- Multiple Availability Zones

Data transfer –

- No charge for inbound data transfer
- Tiered charges for outbound data transfer

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Also consider the number of input and output requests that are made to the database.

Consider the deployment type. You can deploy your DB instance to a single Availability Zone (which is analogous to a standalone data center) or to multiple Availability Zones (which is analogous to a secondary data center for enhanced availability and durability). Storage and I/O charges vary, depending on the number of Availability Zones that you deploy to.

Finally, consider data transfer. Inbound data transfer is free, and outbound data transfer costs are tiered.

Depending on the needs of your application, it’s possible to optimize your costs for Amazon RDS database instances by purchasing Reserved Instances. To purchase Reserved Instances, you make a low, one-time payment for each instance that you want to reserve. As a result, you receive a significant discount on the hourly usage charge for that instance.

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Recorded demo: Amazon RDS console

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Now, take a moment to watch the Amazon RDS console demonstration. The demonstration shows how to perform the following tasks using the AWS Management Console:

- Configure an Amazon RDS installation running the MySQL database engine.
- Connect to the database using a MySQL client.

You can find this video within the module 8 section of the course with the title: **Console Demonstration - RDS** . If you are unable to locate this video demonstration please reach out to your educator for assistance.

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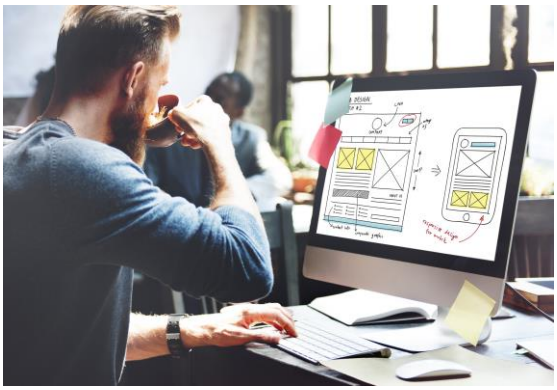
Build Your DB Server and Interact with Your DB Using an App



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You will now complete Lab 5: Build Your DB Server and Interact with Your DB Using an App.

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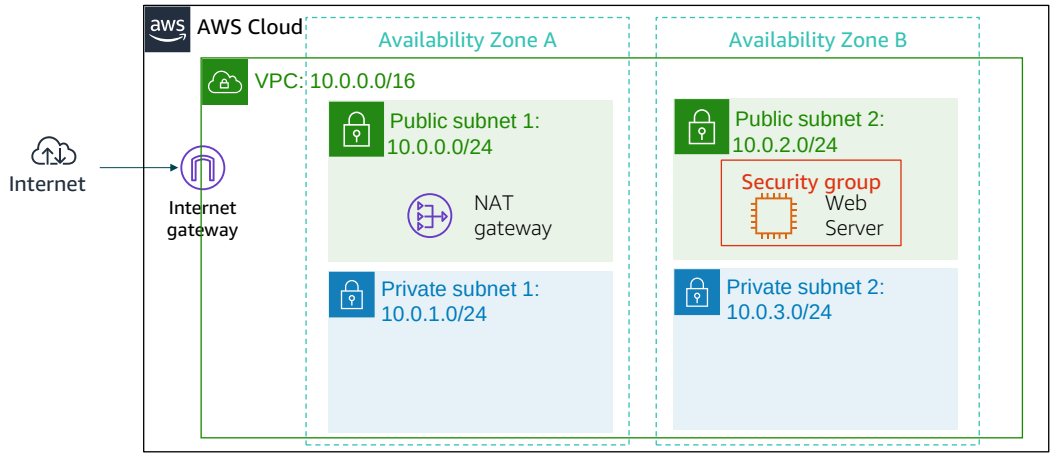
Lab 5: Scenario

This lab is designed to show you how to use an AWS managed database instance to solve a need for a relational database.



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This lab is designed to show you how to use an AWS managed database instance to solve a need for a relational database. With Amazon RDS, you can set up, operate, and scale a relational database in the cloud. It provides cost-efficient and resizable capacity while managing time-consuming database administration tasks, which enables you to focus on your applications and your business. Amazon RDS provides six familiar database engines to choose from: Amazon Aurora, Oracle, Microsoft SQL Server, PostgreSQL, MySQL, and MariaDB.

Amazon RDS Multi-AZ deployments provide enhanced availability and durability for DB instances, which make them a good fit for production database workloads. When you provision a Multi-AZ DB instance, Amazon RDS automatically creates a primary DB instance and synchronously replicates the data to a standby instance in a different Availability Zone.

After completing this lab, you should be able to:

- Launch an Amazon RDS DB instance with high availability.
- Configure the DB instance to permit connections from your web server.
- Open a web application and interact with your database.

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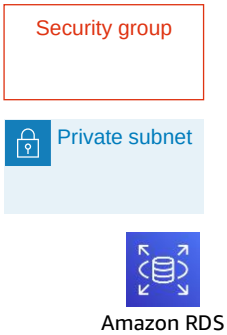
Lab 5: Tasks



Create a **VPC security group.**

Create a **DB subnet group.**

Create an **Amazon RDS DB** instance and interact with your database.

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Your goal in completing this lab is to:

- Create a VPC security group.
- Create a DB subnet group.
- Create an Amazon RDS DB instance and interact with your database.

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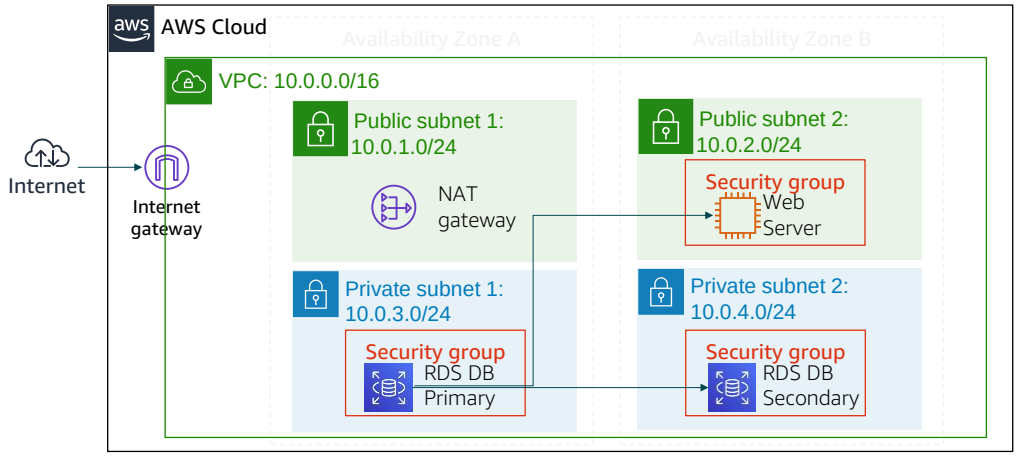
Lab 5: Final product



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In this lab, you:

- Launched an Amazon RDS DB instance with high availability.
- Configured the DB instance to permit connections from your web server.
- Opened a web application and interacted with your database

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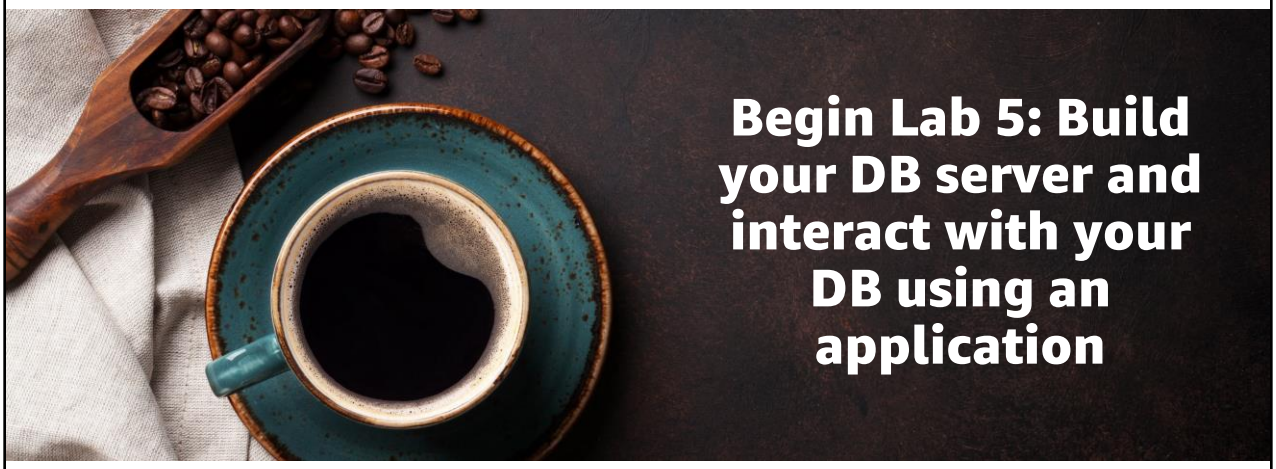
~ 30 minutes



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It is now time to start the lab.

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Lab debrief: key takeaways

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In this lab you:

- Created a VPC security group.
- Created a DB subnet group.
- Created an Amazon RDS DB instance
- Interacted with your database

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Section 1 key takeaways

- With Amazon RDS, you can set up, operate, and scale relational databases in the cloud.
- Features –
  - Managed service
  - Accessible via the console, AWS Command Line Interface (AWS CLI), or application programming interface (API) calls
  - Scalable (compute and storage)
  - Automated redundancy and backup are available
  - Supported database engines:
    - Amazon Aurora, PostgreSQL, MySQL, MariaDB, Oracle, Microsoft SQL Server

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Amazon RDS is a web service that makes it easy to set up, operate, and scale a relational database in the cloud. It provides cost-efficient and resizable capacity while managing time-consuming database administration tasks so you can focus on your applications and your business. Features include that it is a managed service, and that it can be accessed via the console, AWS Command Line Interface (AWS CLI), or application programming interface (API) calls. Amazon RDS is scalable for compute and storage, and automated redundancy and backup is available. Supported database engines include Amazon Aurora, PostgreSQL, MySQL, MariaDB, Oracle, and Microsoft SQL Server.

Amazon RDS supports demanding database applications. You can choose between two solid state drive (SSD)-backed storage options: one option is optimized for high-performance Online Transactional Processing (OLTP) applications, and the other option works well for cost-effective, general-purpose use.

With Amazon RDS, you can scale your database's compute and storage resources with no downtime. Amazon RDS runs on the same highly reliable infrastructure that is used by other AWS services. It also enables you to run your database instances and Amazon VPC, which is designed to provide you with control and security.

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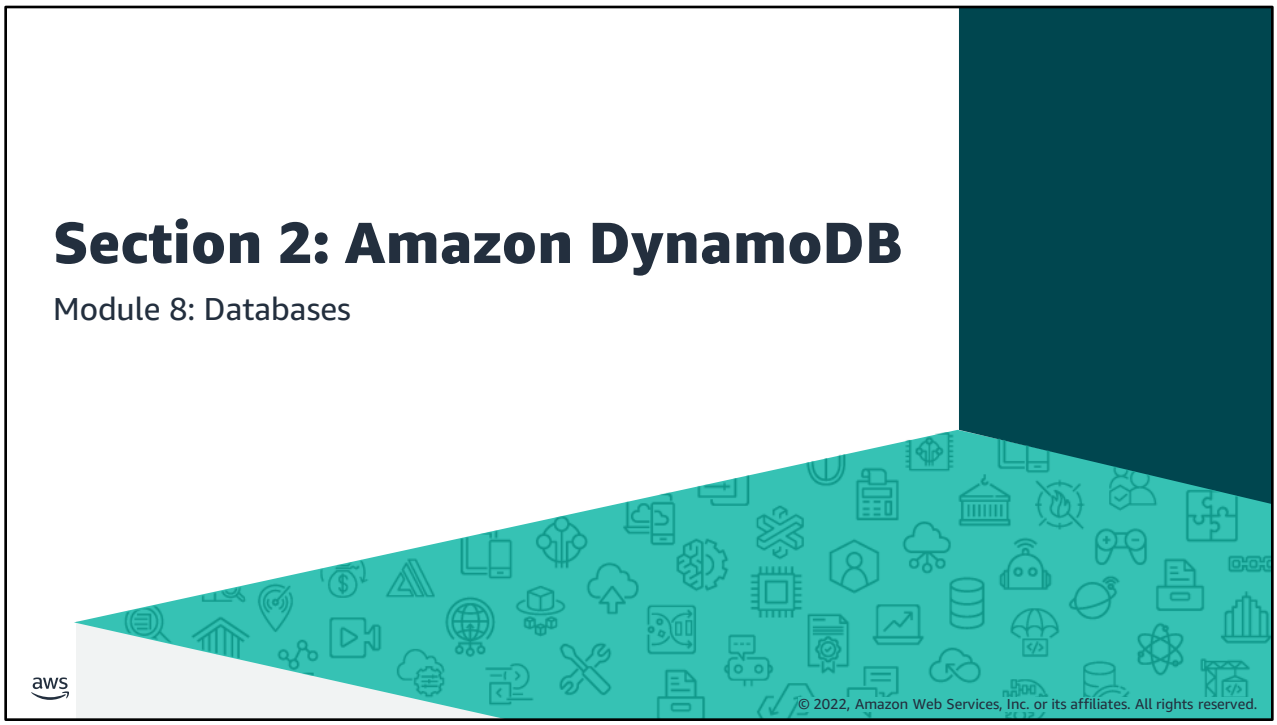

Welcome to Section 2: Amazon DynamoDB.

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